DATA ITEM DESCRIPTION

Title: VIBRATION AND HIGH CYCLE FATIGUE ANALYSIS REPORT

Number: DI-ENVR-82491
Approval Date: 20250709
AMSC Number: F10577
Limitation: N/A
DTIC Applicable: No
GIDEP Applicable: No
Preparing Activity: 11 (AFLCMC/EZFS)
Project Number: ENVR-2025-011
Applicable Forms: N/A

Use/Relationship: The Vibration and High Cycle Fatigue Analysis Report describes the approach required for establishing the vibration environments of the structure and equipment and predicting the strength and fatigue life of airframe structures subjected to these environmental loads. This analysis report supports the design of the air vehicle.

a. This Data Item Description (DID) is applicable to any contract for which an air vehicle is under development, a modification is made, or an Environmental Criteria Document (ECD) is required.

b. This DID contains the format, content, and intended use information for the data deliverable resulting from the work task described in section 5 of MIL-STD-1530, *Aircraft Structural Integrity Program (ASIP)* (copies of this document are available online at https://quicksearch.dla.mil).

c. This DID supersedes DI-ENVR-82121, *Vibration and Acoustic Analyses Reports – Air Vehicle* (copies of this document are available online at https://quicksearch.dla.mil).

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. The Vibration and High Cycle Fatigue Analysis Report shall be in the contractor's format.

3. Content. The Vibration and High Cycle Fatigue Analysis Report shall describe the contractor's analysis associated with the design for vibration environments and shall include the following:

a. Vibration Loads and Environmental Analysis, to include:

(1) Approach used to establish the vibration loading and environmental exposure of structure and equipment for ground and flight operations.

(2) Descriptions of all sources of vibration resulting from the operation of propulsion and other power systems, gusts, wakes, oscillating shocks, and other aerodynamic disturbances, Mach effects, ground operations, missile and rocket launches, and gun fire.

(3) Substantiation of any design provisions employed to control vibration load levels.

(4) Predicted levels for harmonic, transient, and random vibration sources or combinations of sources with graphs, calculations, data tables, etc., in sufficient detail to derive the exposure of the system by physical location and with sufficient frequency resolution to evaluate local design features.

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(5) Description of the vibration loading with graphs and tables that define the loading as a function of frequency and exposure time.

(6) Relevant service environment details that contribute to the vibration loading and resulting service life, including temperature, humidity levels, and cabin pressurization.

(7) Description of analysis tools and models.

(8) Usage data in terms of mission types, mission mix, storage, and transportation used to develop the repeated loads and environmental spectra for use in the service life analysis.

(9) Descriptions of the theories, methods, and assumptions used to modify vibration levels to support the accelerated testing procedures.

b. Fatigue Analysis, to include:

(1) High cycle fatigue analysis methodology and validation results used to design the air vehicle structure and equipment subjected to vibration loading.

(2) Description of analysis tools and analytical models.

(3) Fatigue life predictions of all vibration sensitive structure and equipment and all supporting calculations and data, such as: loads-to-stress transfer functions, material properties, and transmissibility factors

(4) Usage data in terms of mission types, mission mix, storage, and transportation used to develop the repeated loads and environmental spectra for use in the service life analysis.

c. Analysis Correlation, to include:

(1) Comparison of pre-test analysis predictions to the vibration, environmental, and high cycle fatigue test data.

(2) Analysis results and correlation of the analytical tools used in the vibration, environmental, and high cycle fatigue analysis.

(3) Comparison of the vibration, environmental, and high cycle fatigue test results with previous design levels and qualification test requirements.

End of DI-ENVR-82491

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